

TransTech Systems

Inventors of Capacitance Technology for Asphalt Density Measurement

TransTech Systems, Inc. develops innovative, nondestructive evaluation instruments for the construction industry. Its products are built around the company's core capacitance-based technology, which enables fast, safe, and practical quality-control measurements in the field. For more than 30 years, TransTech Systems has advanced the use of capacitance technology in road construction. **Through continuous development and refinement, the company has become a recognized leader in non-nuclear quality-control technology for determining in-process asphalt density.**



**PQI
Prototype
1996**



**PQI
100
1998**



**PQI
200
1999**



**PQI
301 Series
2001-2004**



**PQI
380
2010**



**PQI
380+
2025**

At TransTech Systems, we are all about continuous development of innovative solutions to difficult problems in a way that seems easy to the users of our devices and helps them save time and money.

At TransTech Systems, innovation means solving difficult field problems in ways that feel simple to the people using our devices. Our goal is to develop practical solutions that help construction professionals work more efficiently, improve quality control, and save time and money.

For many years, nuclear density gauges were the standard method for quality-control testing on road construction projects. The founders of TransTech Systems, drawing on extensive industry experience, recognized that many construction and civil engineering practices had changed very little over time.

More than 25 years ago, TransTech Systems introduced a new approach: applying established electrical technology to characterize asphalt during construction and determine density in seconds. This work led to patented technology and the first prototype for longitudinal joint density testing, which became the Pavement Quality Indicator, or PQI 100.



1-800-724-6306
transtechsys.com



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The PQI quickly demonstrated the value of non-nuclear asphalt density measurement and gained momentum across the construction industry. The speed of test, consistency of test results and improved safety for operators supported the widespread use of this innovative technology. Since the first commercial unit was introduced, TransTech Systems has continued improving its flagship product to increase accuracy, reliability, ruggedness, and ease of use across a wide range of mixes and jobsite conditions.



The latest fifth-generation PQI Model 380+ builds on this history with features designed to make field testing faster, easier, and more connected. A large LCD touchscreen has been added for an improved user interface. Integrated W-Fi and Bluetooth capability allows immediate data transfer by uploading density test results from the gauge directly to TransTech Connect.

TransTech Connect is a cloud based quality control platform developed exclusively for TransTech Systems' non-nuclear density gauges. When paired with the PQI 380+, it streams density readings from the mat to the office as they are taken, helping teams access field data more quickly and manage measurement information more efficiently. TransTech Connect is the first system in the asphalt industry to combine fully non-nuclear electromagnetic density sensing with live cloud synchronization.

The PQI is used worldwide, supported by distributors in more than 87 countries. Together with TransTech Systems' Soil Density Gauge, thousands of units are in use every day. This global presence gives the company valuable insight into real-world jobsite conditions across diverse climates, materials, and applications, insight that continues to guide product development and technology improvements.



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